CUAI HAR 스터디

2025.11.17

발표자 : 방세현

스터디원 소개 및 만남 인증

스터디원 1 : 권순범

스터디원 2 : 남승운

스터디원 3 : 방세현

스터디원 4 : 정현석

스터디 일정

중간고사 ~ 기말고사

- 목표: HAR 맛보기

- 내용:

내용	10주차	11주차	12주차	13주차	14주차	15주차
HAR 리뷰 논문 읽기						
tutorial 돌리면서 몸으로 익히기						
HAR 다른 논문 각자 리뷰						
논문 발표						
논문 구현 (선택)						

Past, Present, and Future of Sensor-based Human Activity Recognition Using Wearables: A Surveying Tutorial on a Still Challenging Task

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In the many years since the inception of wearable sensor-based Human Activity Recognition (HAR), a wide variety of methods have been introduced and evaluated for their ability to recognize activities. Substantial gains have been made since the days of hand-crafting heuristics as features, yet, progress has seemingly stalled on many popular benchmarks, with performance falling short of what may be considered 'sufficient'–despite the increase in computational power and scale of sensor data, as well as rising complexity in techniques being employed. The HAR community approaches a new paradigm shift, this time incorporating world knowledge from foundational models. In this paper, we take stock of sensor-based HAR – surveying it from its beginnings to the current state of the field, and charting its future. This is accompanied by a hands-on tutorial, through which we guide practitioners in developing HAR systems for real-world application scenarios. We provide a compendium for novices and experts alike, of methods that aim at finally solving the activity recognition problem.

CCS Concepts: • **Human-centered computing → Human computer interaction (HCI); Ubiquitous and mobile computing**; • **Computing methodologies → Machine learning.**

Paper Review 일부 소개

History: from handcrafted features to DNNs

Feature Engineering and the Activity Recognition Chain (ARC)

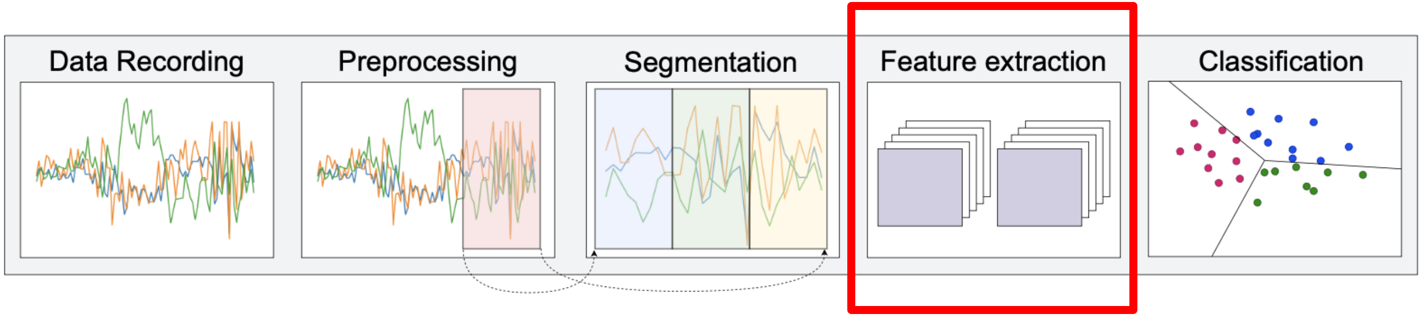
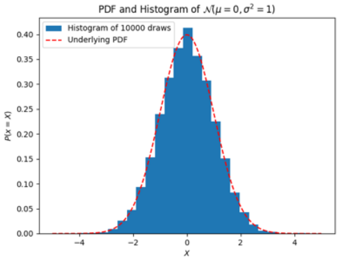


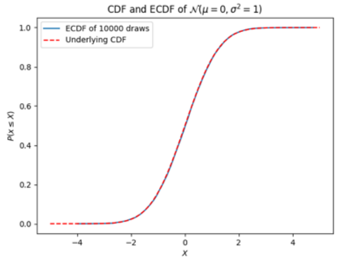
Fig. 1. The Activity Recognition Chain as summarized in [19].

History: from handcrafted features to DNNs

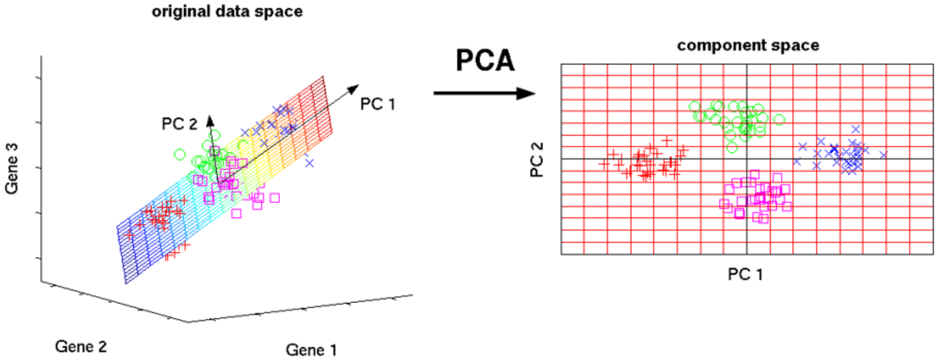
(초기)

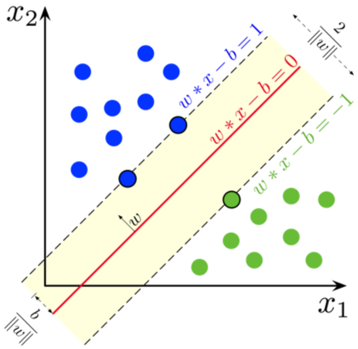
- handcrafted
- statistical
- distribution-based features
- Principle Component Analysis (PCA)





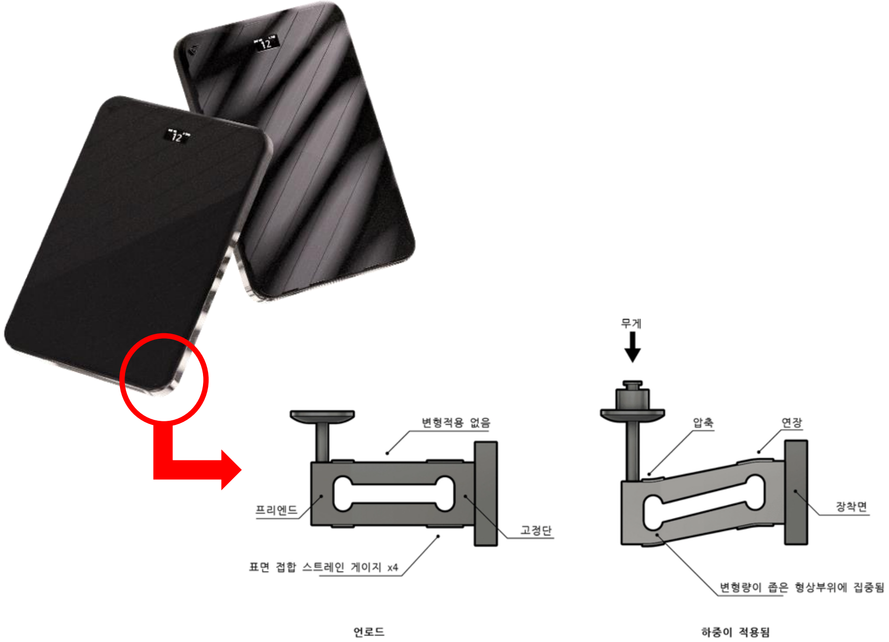
- classical machine learning approaches (SVM, logistic regression, kNNs, or Random Forest)

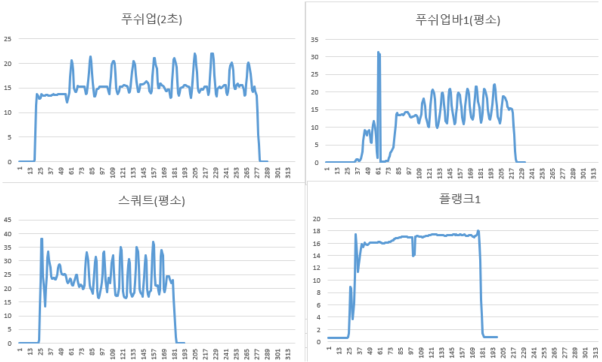




History: from handcrafted features to DNNs

(초기 / Rule base Feature & ML의 어려움 - 예)

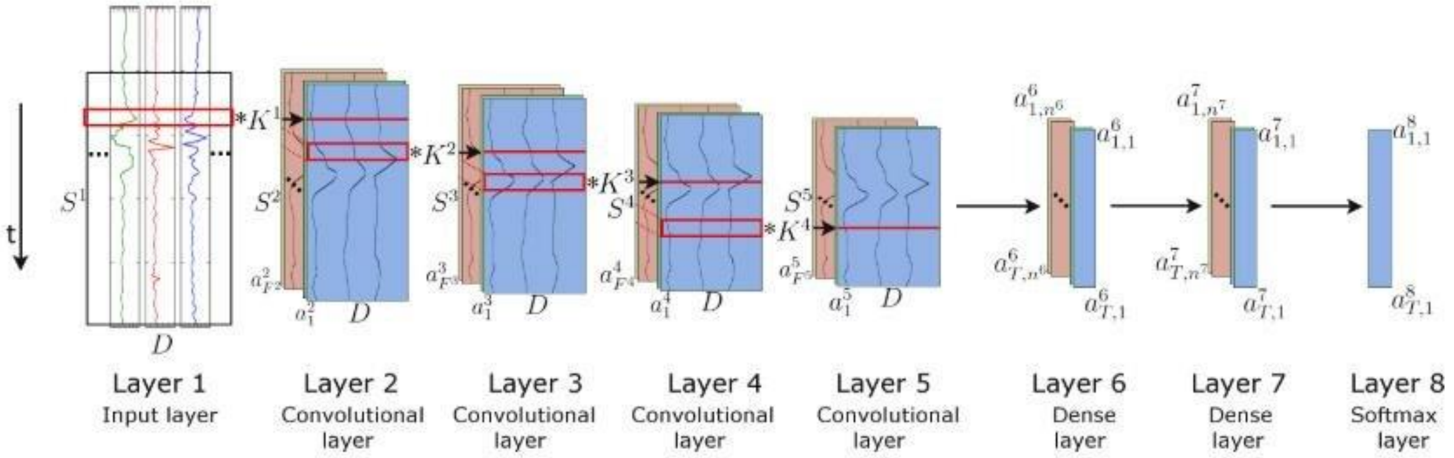




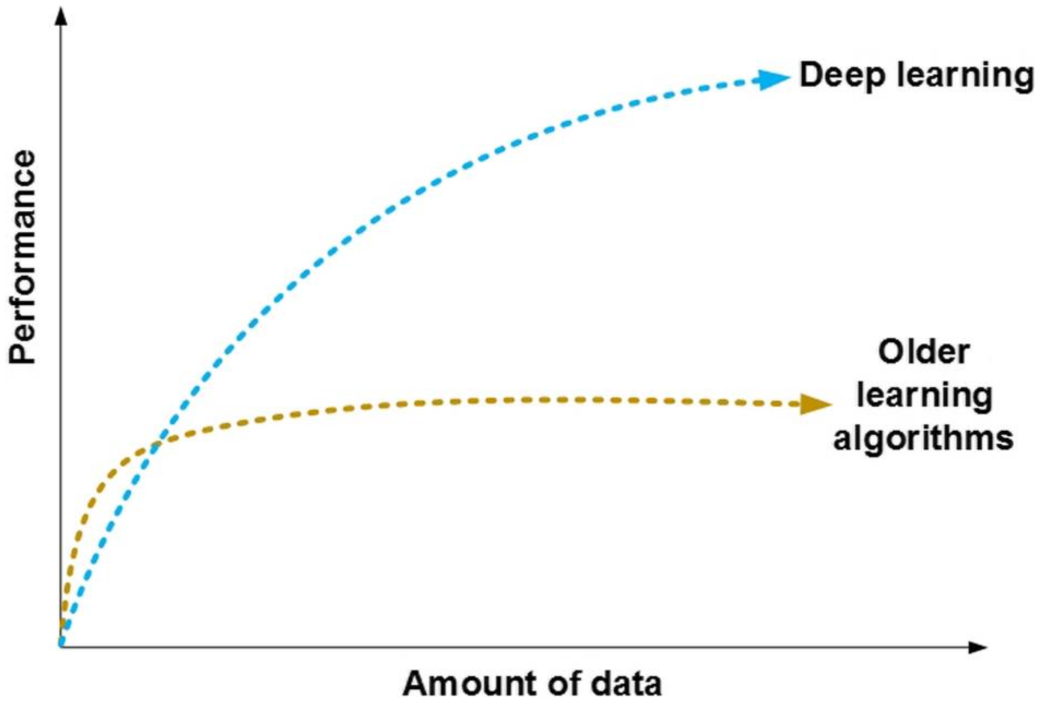
History: from handcrafted features to DNNs

(최근)

- Deep Learning Based (e.g. DeepConv LSTM)

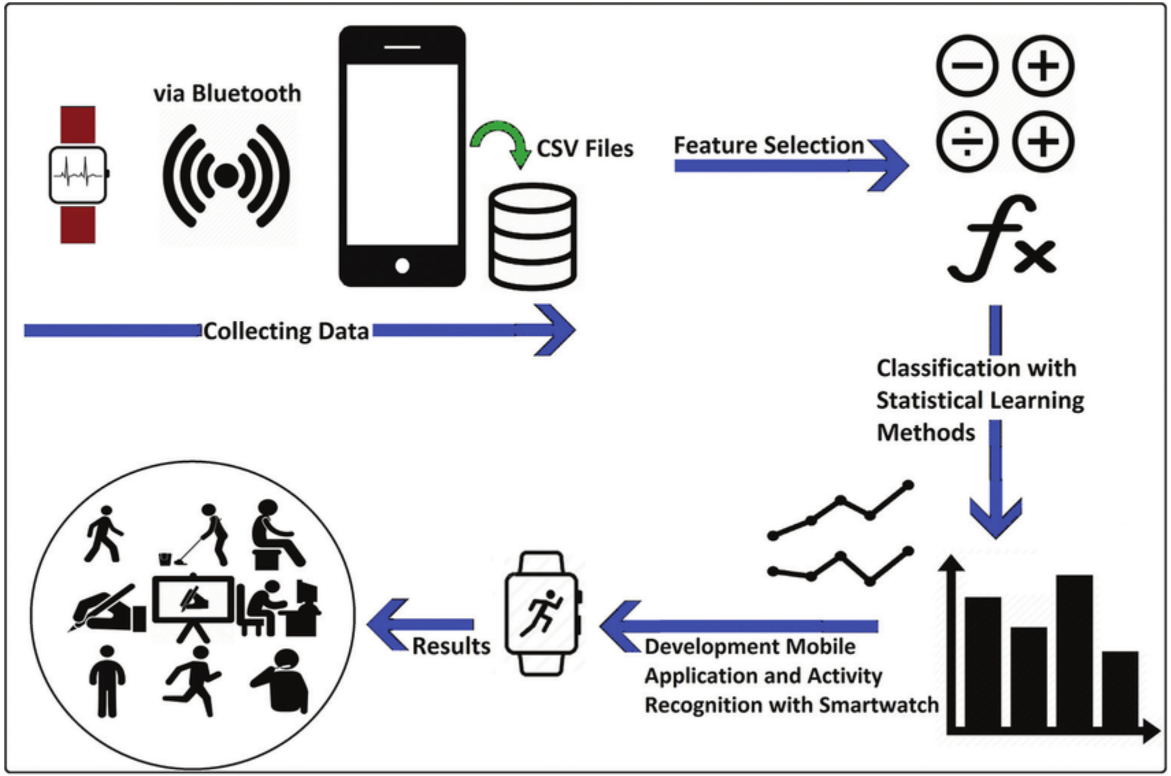


The Importance of Representations



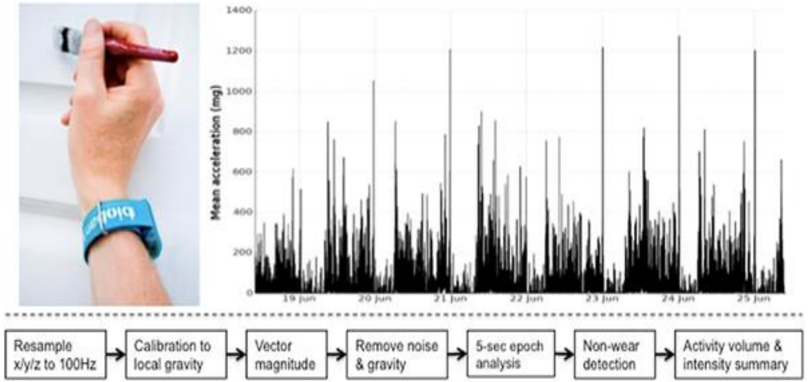
The Importance of Representations

근데 labeled data가 없다...



The Importance of Representations

하지만, unlabeled은 있죠?!



	Wear time [median (IQR) hours]		Acceleration vector magnitude [mean +- stdev mg]	
	Women	**Men**	**Women**	**Men**
Age (yrs) [A]				
45–54	164.9 (152.4–167.0)	165.4 (149.5–168.0)	31.2 +- 8.7	31.1 +- 9.7
	(n = 12,586)	(n = 8655)	(n = 11,572)	(n = 7838)
55–64	165.4 (156.0–167.0)	165.8 (156.5–168.0)	29.1 +- 8.0	28.8 +- 8.8
	(n = 21,322)	(n = 14,410)	(n = 19,890)	(n = 13,362)
65–74	165.6 (159.1–168.0)	166.8 (160.8–168.0)	26.6 +- 7.1	25.6 +- 7.7
	(n = 22,821)	(n = 20,595)	(n = 21,489)	(n = 19,385)
75–79	165.6 (158.9–167.0)	166.8 (162.6–168.0)	23.9 +- 6.5	22.9 +- 6.8
	(n = 1494)	(n = 1695)	(n = 1436)	(n = 1628)
p value	p<0.001	p<0.001	p<0.001	p<0.001
Time of day [B]				
0–5.59 $_{am}$	40.9 (36.0–42.0)	42.0 (36.6–42.0)	4.4 +- 3.1	4.9 +- 4.4
	(n = 58,223)	(n = 45,355)	(n = 54,387)	(n = 42,213)
6–11.59 $_{am}$	41.0 (38.9–42.0)	42.0 (39.0–42.0)	38.6 +- 14.9	37.4 +- 16.4
	(n = 58,223)	(n = 45,355)	(n = 54,387)	(n = 42,213)
12–5.59 $_{pm}$	42.0 (40.3–42.0)	42.0 (40.3–42.0)	44.3 +- 13.8	42.9 +- 16.0
	(n = 58,223)	(n = 45,355)	(n = 54,387)	(n = 42,213)
6–11.59 $_{pm}$	42.0 (39.5–42.0)	42.0 (40.2–42.0)	26.4 +- 10.4	24.9 +- 11.5
	(n = 58,223)	(n = 45,355)	(n = 54,387)	(n = 42,213)
p value	p<0.001	p<0.001	p<0.001	p<0.001
Day [C]				
Weekday	23.7 (22.5–24.0)	23.8 (22.6–24.0)	28.5 +- 8.2	27.5 +- 9.0
	(n = 58,223)	(n = 45,355)	(n = 54,387)	(n = 42,213)
Weekend	24.0 (22.9–24.0)	24.0 (23.3–24.0)	28.0 +- 9.4	27.1 +- 10.8
	(n = 58,223)	(n = 45,355)	(n = 54,387)	(n = 42,213)
p value	p<0.001	p<0.001	p<0.001	p<0.001
Season [D]				
Spring	165.6 (156.2–167.5)	166.1 (157.4–168.0)	28.8 +- 8.0	28.1 +- 9.1
	(n = 13,365)	(n = 10,224)	(n = 12,480)	(n = 9,469)
Summer	165.4 (156.2–168.0)	166.3 (157.4–168.0)	28.8 +- 8.1	28.2 +- 8.7
	(n = 15,450)	(n = 11,943)	(n = 14,353)	(n = 11,016)
Autumn	165.6 (157.2–167.0)	166.3 (158.2–168.0)	28.3 +- 8.0	27.3 +- 8.7
	(n = 17,213)	(n = 13,506)	(n = 16,157)	(n = 12,633)
Winter	165.6 (156.7–168.0)	166.3 (157.9–168.0)	27.7 +- 7.8	26.3 +- 8.4
	(n = 12,195)	(n = 9,682)	(n = 11,397)	(n = 9,095)
p value	p = 0.289	p = 0.104	p<0.001	p<0.001

[A] *Age*: Kruskal-Wallis test used to compare wear-time distributions, and one-way analysis of variance test used to compare acceleration vector magnitude means. Sum wear time hours for week displayed (max = 168.0).
[B] *Time of day*: Friedman test used to compare wear-time distributions within individuals, and repeated one-way analysis of variance test used to compare acceleration vector magnitude means within individuals and between age groups. Sum wear time hours for time quadrant over a week displayed (max = 168.0).
[C] *Day*: Wilcoxon test used to compare wear-time distributions within individuals, and repeated one-way analysis of variance test used to compare acceleration vector magnitude means within individuals and between age groups. Average wear time hours for day displayed (max = 24.0).
[D] *Season (Spring starting on 1st March)*: Kruskal-Wallis test used to compare wear-time distributions, and two-way analysis of variance test used to compare acceleration vector magnitude means between age groups. Sum wear time hours for week displayed (max = 168.0).

doi:10.1371/journal.pone.0169649.t001

- **9만명**
- **20 TB of sensor data**

Aiden Doherty, Dan Jackson, Nils Hammerla, Thomas Plötz, Patrick Olivier, Malcolm H Granat, Tom White, Vincent T Van Hees, Michael I Trenell, Christoper G Owen, et al. 2017. Large scale population assessment of physical activity using wrist worn accelerometers: the UK biobank study. PloS one 12, 2 (2017), e0169649.

The Importance of Representations

Pretrain-then-finetune

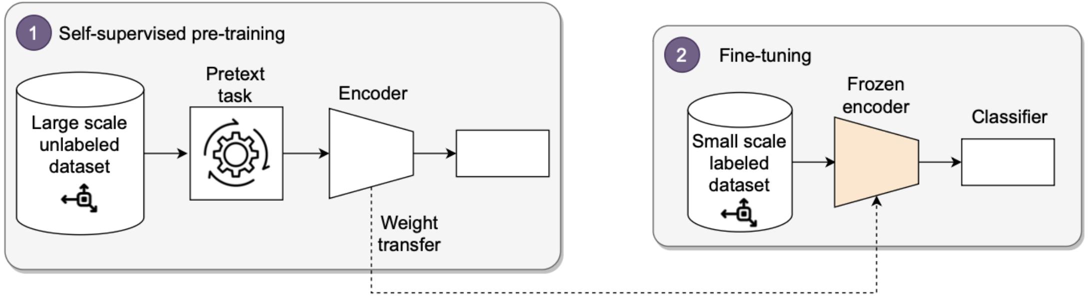
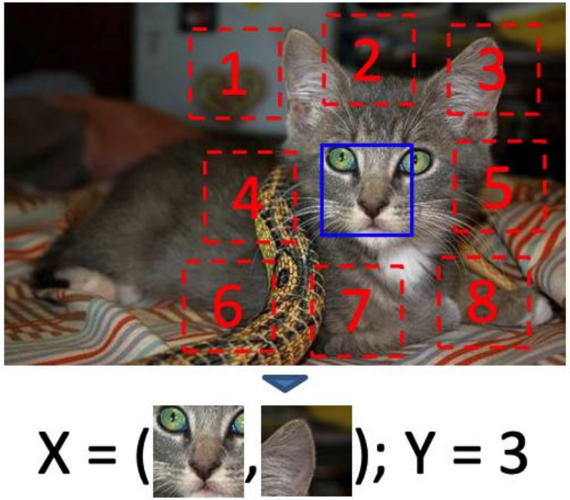


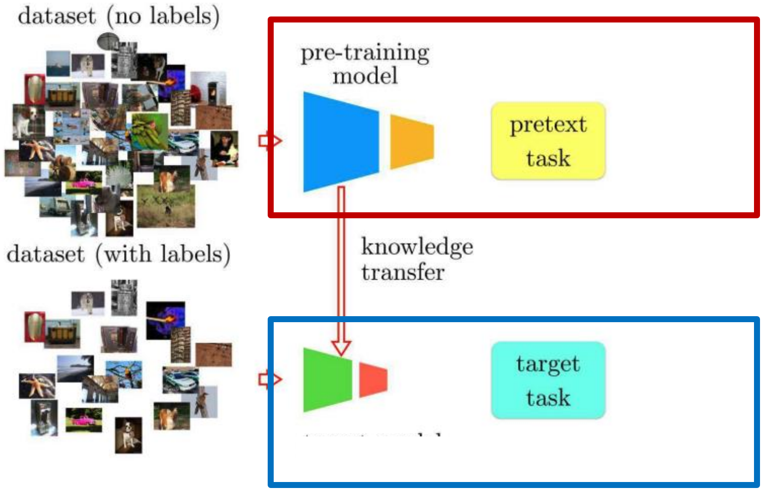
Fig. 2. An overview of the self-supervised learning pipeline. Reproduced with permission from [77].

The Importance of Representations

Pretrain?

self-supervised learning first solves a **pretext task**, and then tackles the **target task**.



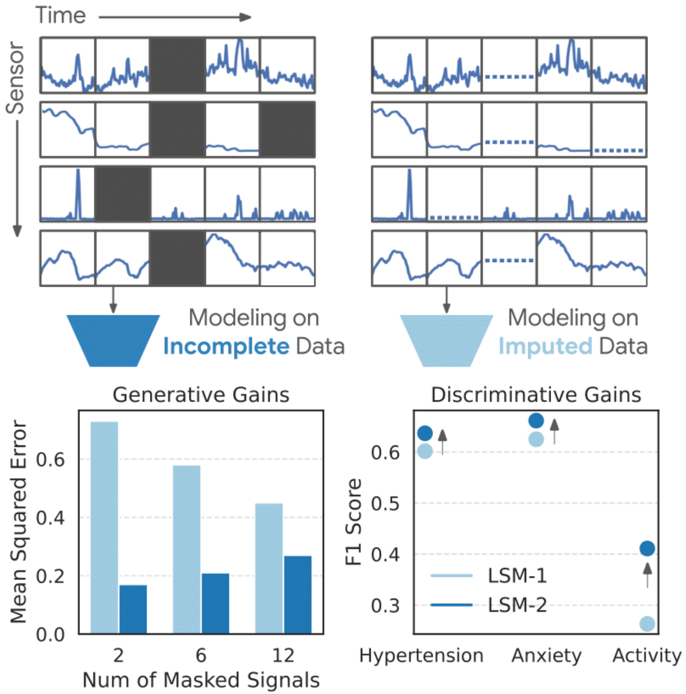


1. Doersch et al., "Unsupervised Visual Representation Learning by Context Prediction," ICCV 2015
2. Noroozi et al., "Boosting Self-Supervised Learning via Knowledge Transfer," CVPR 2018.

Self-Supervised Representation Learning

SSL 종류

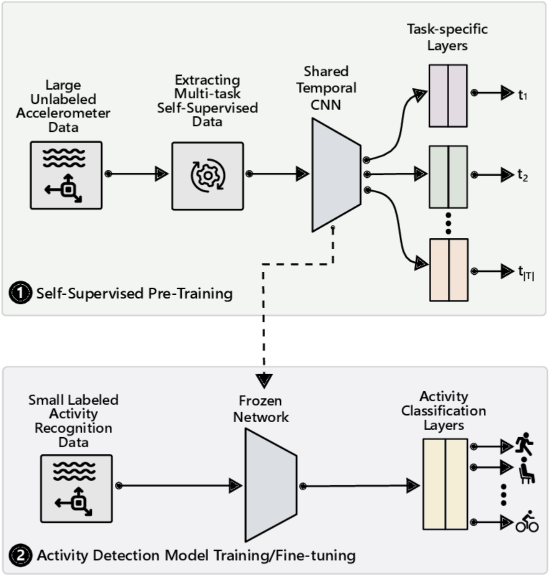
- Masked Reconstruction



Self-Supervised Representation Learning

SSL 종류

- Multi-Task SSL
  - **하나의 모델이 여러 개의 다른 작업(task) 을 동시에 학습**
  - **목표:** 서로 다른 작업을 동시에 수행하면서, 모델이 더 일반화되고 풍부한 특징을 학습



Representation Alignment and Structuring

자 이제,

- **Label 이 없는 대규모 data를 활용**하기 위한 **SSL** 을 알았다.
- SSL은 레이블이 없는 대규모 데이터를 활용하여 representations을 사전 학습하는 'pretrain-then-finetune'

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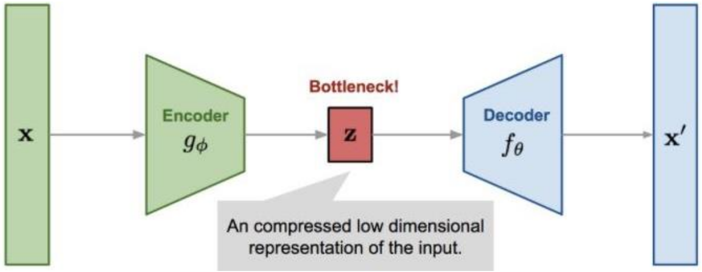
How? Representation alignment?



Representation Alignment and Structuring

- **Contrastive Learning의 등장!**

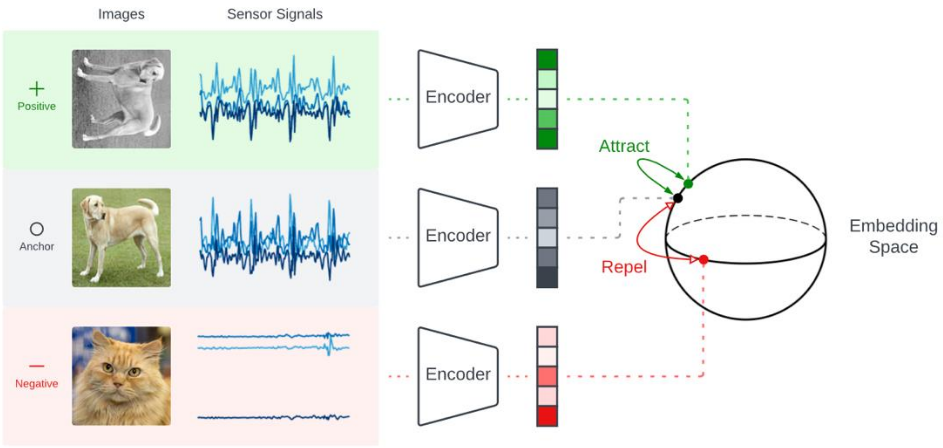
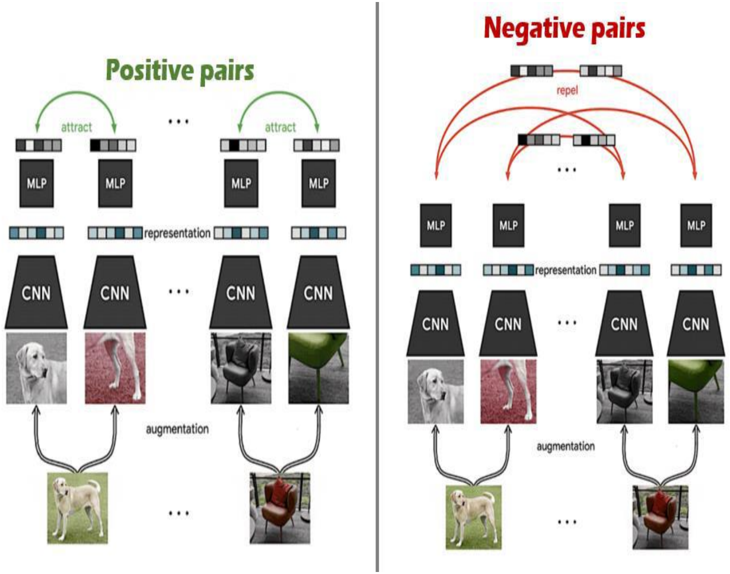


Fig. 3. An overview of contrastive learning (CL). Data samples (images or sensor signals depending on the target task) are first passed through an encoder to obtain embeddings in a latent space. These embeddings are pushed closer to each other or pulled apart depending on whether the embedding forms a positive pair or a negative pair with the anchor.

Representation Alignment and Structuring

- **Contrastive Learning?**
- *The model learns to **make representations of images from the same source similar**, and **representations of images from different sources as dissimilar as possible**.*



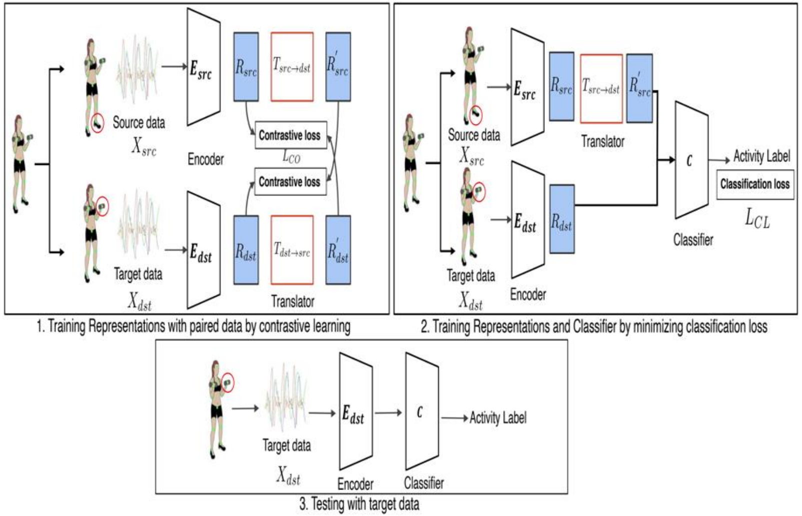


Figure 1: Overall network architecture and steps of the proposed method.

지금까지 HAR 역사에 대한 간략한 개괄이었고,

센서 처리에 대한 최신 방법론들은

남은 논문을 읽어보시길 추천드립니다!

감사합니다!